FACULTY OF ENGINEERING

QUALITY ASSURANCE: TANGRA INC.

INGB413

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1. Background

1.1. Introduction to Quality Management in Manufacturing

Over the years, quality management has emerged as a key factor guiding organisational performance. It is defined as a structured scientific approach that integrates quality into all organisational processes, from product design to delivery, with the aim of meeting customer requirements while improving operational efficiency (Lim, 2019:11). Quality management involves a combination of coordinated actions that direct and control an organisation's quality efforts (Dale, Papalexi, Bamford & Van der Wiele, 2016:3). Foley (2004:85) describes quality management as a revolutionary management philosophy, offering a new mindset for managing organisations, a paradigm shift, an alternative to control-based management, and a framework for enhancing overall performance and competitiveness.

Studies show that reducing variability in materials, methods, and people through quality management enhances product conformity and customer satisfaction, leading to better organisational outcomes (Misra, 2008:164). Powell (1995:1204) found that Total Quality Management (TQM) provides a source of sustainable competitive advantage. Nair (2006:951) also identified a positive correlation between quality management practices and firm performance, while Kasie, Belay, and Helo (2014:166) demonstrated that quality management practices are associated with increased labour productivity. Alzoubi, Ahmed, and Al-Azzam (2019:317) affirm that TQM practices significantly improve customer satisfaction, market competitiveness and overall firm performance. Tan, Kannan, and Narasimhan (2007) emphasise that organisations must continually improve quality and productivity to meet customer demands and sustain profitability.

Quality management holds particular importance for Small and Medium-sized Enterprises (SMEs), which represent approximately 99% of global businesses (Gilmore, Carson and Grant, 2001:6). SMEs play a crucial role in promoting economic growth, job creation, and innovation. In engineering manufacturing, quality management is not merely a regulatory formality; it is fundamental to achieving precision, cost control, and global competitiveness (Oakland, 2014:4; Madanhire & Mbohwa, 2016:580).

1.2. Problem statement: Tangra Inc. Case Study

Tangra Inc., a small-scale manufacturer of tangram puzzles, is experiencing an increase in defective products within its assembly line, which hinders its ability to meet market demand and maintain profitability. Quality management, defined as a philosophy built on core principles and supported by specific practices and techniques, is essential to addressing such challenges (Dean and Bowen, 1994:395).

Defect reduction in manufacturing requires systematic methods for identifying and address defects throughout the product lifecycle, aiming to improve the overall quality and reliability of the product, leading to increased customer satisfaction, reduced costs, and improved efficiency (Sadikin, 2023:75). Defective products impose significant financial burdens, which necessitate techniques such as Statistical Process Control (SPC), root cause analysis, and process improvement (Montgomery, 2020:23). SPC, distinguishes between normal process variation and assignable causes, enabling timely corrective action before defects occur (Montgomery, 2013:130; Wolniak and Grebski, 2023:260). Godina, Pimentel and Matias (2018:8) emphasise the potential of SPC for improving defect detection and process stability, while Teplická and Hurná (2021:22) advocate for the use of the Prevention-Appraisal-Failure (PAF) model to monitor quality costs and improve SME efficiency.

The report investigates two defect detection processes at Tangra Inc. Process 1 involves batch sampling and defect removal before assembly; Process 2 involves real-time immediate detection and immediate discarding of defective assemblies during production. The objective is to determine which process is more effective in reducing defects, improving product quality, and supporting Tangra Inc.'s operational goals.

1.3. Objectives

The main objectives of this study are to:

- To evaluate the effectiveness of process 1 (batch sampling) and process 2 (real-time defect discarding) in reducing defect rates.
- To quantify the cost implications of each process and assess overall profitability.
- To analyse the potential of defect reduction strategies to improve Tangra Inc.'s competitiveness in the manufacturing sector.

Defect reduction, a key objective of quality management, involves systematic approaches to identifying, analysing, and addressing defects throughout the product lifecycle. It aims to improve product reliability, reduce costs, and increase customer satisfaction (Sadikin, 2023:75).

1.4. Overview of the Experimental Process:

An experimental study was conducted at Tangra Inc. to investigate the increase in defective puzzle pieces. Two processes were tested, each for a 15-minute interval. Data collected included the number of defects, the number of completed tangrams and related operational details.

1.4.1. Process 1: Batch sampling and defect removal

In process 1, raw materials are received in bulk. Samples of five puzzle pieces (n=5) are drawn systematically and inspected for defects. Any defective parts are removed before assembly proceeds. Any defective parts are removed before assembly proceeds. This approach reflects a traditional quality control method, where inspection and defect removal occur before the main assembly phase, based on a predetermined sample size (Montgomery, 2020:87).

1.4.2. Process 2: real-time defect identification and immediate discarding

Process 2 uses pre-sorted raw materials, stored in colour-coded bins at the assembly stations. Assemblers select parts directly from these bins. If a defect is detected during assembly, the entire semi-assembled tangram, including the wooden board and placed pieces, is discarded immediately. For each discarded unit, the number of placed pieces and the number of defective parts is recorded. This method represents a real-time, in-process quality control approach, where defect discovery triggers immediate rejection of the assembly.

1.5. Cost Implications

Cost implications of these processes are significant and directly influence profitability. Each puzzle piece costs R2.00, and each wooden board costs R5.00. A defective semi-assembled item results in a loss of R20.00. Each completed tangram product was sold for R25.00. Therefore, comparing the two processes must account for both the quality outcomes and the overall profitability of Tangra Inc.

2. Literature review

The literature review examines major themes, methodologies, theoretical frameworks, and anthropological implications of quality management, with a particular focus on Statistical Process Control (SPC), and identifies gaps in the literature pertinent to Small and Medium Enterprises (SMEs) in the South African manufacturing context, as well as Tangra Inc.

2.1. Major themes and methodologies in Quality Management in Manufacturing

Quality management in manufacturing has evolved from inspection-focused practices to holistic systems that emphasise continuous improvement, defect prevention, and customer satisfaction. Early work by Shewhart in the 1920s pioneered control charts to monitor process variability (Montgomery, 2020:23), which have cemented the foundation for Statistical Process Control (SPC) as a management tool. Deming’s post-war interventions in Japan transformed

SPC into a management philosophy, linking process stability to broader organisational success (Godina, Pimentel and Matias, 2018:4).

In the South African context, SMEs face unique challenges in implementing QM systems because of resource constraints, limited technical expertise, and infrastructural limitations. SMEs often struggle with the complexity and cost associated with these systems (Magodi, Daniyan, and Mpofu, 2023:5).

2.2. Theoretical Frameworks and perspectives

Quality capabilities, such as SPC proficiency, act as strategic resources that create competitive advantage (Barney, 1991:103). Systems theory frames the organisation as an interrelated network of subsystems, where quality outcomes depend on feedback loops and systemic alignment (Checkland, 1999:47). Godina, Pimentel and Matias (2018:9) emphasise that leadership commitment and organisational culture are important for quality practices. Sadikoglu and Zehir (2010:403) argue that employee empowerment and cross-functional collaboration are central to successful defect reduction. Leadership commitment and employee involvement are important factors that influence the effectiveness of QM systems (Bhasin, 2012:124) These insights suggest that quality management is not merely technical but deeply social, shaped by cultural norms, communication patterns, and leadership behaviours.

2.3. Statistical Process Control: A critical analysis

Statistical process control is a methodological approach that utilises statistical tools to monitor and control processes, aiming to ensure product quality and process stability. Developed by Shewhart in the 1920s, SPC focuses on identifying and reducing variability within processes (Montgomery, 2020:23). The key tools include control charts, process capability analysis and design of experiments.

Quantitatively, SPC has reduced process variability and continues to enhance product quality. Görmen (2022:648) states that the application of SPC in gear wheel production led to significant improvements in process stability and product consistency. Process capability indices provide measurable indicators of process performance, facilitating data-driven decision-making (Montgomery, 2020:47). On a qualitative perspective, the successful implementation of SPC is influenced by organisational culture, employee training, and management support. In South African SMEs, challenges such as limited statistical expertise, resistance to change, and inadequate training hinder the effective adoption of SPC (Magodi, Daniyan, and Mpofu, 2023:7). According to Madanhire and Mbohwa (2016:583), the complexity of SPC tools can be daunting for SMEs which lack specialised personnel.

In the South African manufacturing sector, the adoption of SPC among SMEs is limited. Factors such as financial constraints, lack of awareness, and insufficient training resources contribute to the low uptake. However, tailored training programs and simplified SPC tools can enhance adoption rates and effectiveness in SMEs (Neo, Mukwakungu, and Lumbwe, 2020:4).

2.4. Comparative Analysis of Defect Reduction strategies

Defect reduction strategies vary in strengths and limitations. Six Sigma offers precision in defect quantification and process improvement through rigorous statistical methods, yet its complexity and resource intensity may be prohibitive for SMEs (Antony, 2006:25). Lean manufacturing excels in eliminating waste and improving flow but may underprioritise defect prevention when market demands fluctuate (Bhasin, 2012:126). Most research focuses on large-scale manufacturers, highlighting a critical gap in empirical evidence (Madanhire and Mbohwa, 2016: 581; Sadikin, 2023: 88).

SPC offers a statistical approach to monitoring and controlling processes, which enables early detection of deviations. While SPC is effective in identifying process issues, its success depends on proper implementation and organizational support. In SMEs, integrating SPC with other methodologies, such as Lean or TQM, can provide a more comprehensive approach to quality improvement (Sreedharan, Sunder, and Raju, 2018:52).

2.5. Gaps in literature

Several gaps persist in quality management relevant to South African SMEs. First, empirical research on defect reduction predominantly centres on large enterprises, marginalising SMEs (Sadikin, 2023:90). Second, the long-term sustainability of defect reduction strategies is not sufficiently explored, as most studies focus on short-term interventions (Kasie, Belay and Helo, 2014:169). Existing studies often overlook the unique challenges faced by these enterprises, such as resource limitations and lack of technical expertise. Future research should explore context-specific strategies for implementing SPC in SMEs, considering factors like simplified tools, tailored training programs, and management support mechanisms (Magodi, Daniyan, and Mpofu, 2023:9).

2.6. Quality management tools

Selecting the appropriate tool depends on the nature of the problem, the data available, and the organisational context (Bhasin, 2012:125). For Tangra Inc., implementing key quality management tools can enhance defect detection, improve process stability, and reduce the costs associated with defective products. ISO 9001 standards have become essential benchmarks for quality management systems, reflecting the increasing recognition of certification as integral to global management practices (Ahmed, 2017; Lourenço et al., 2012). The following tools are recommended:

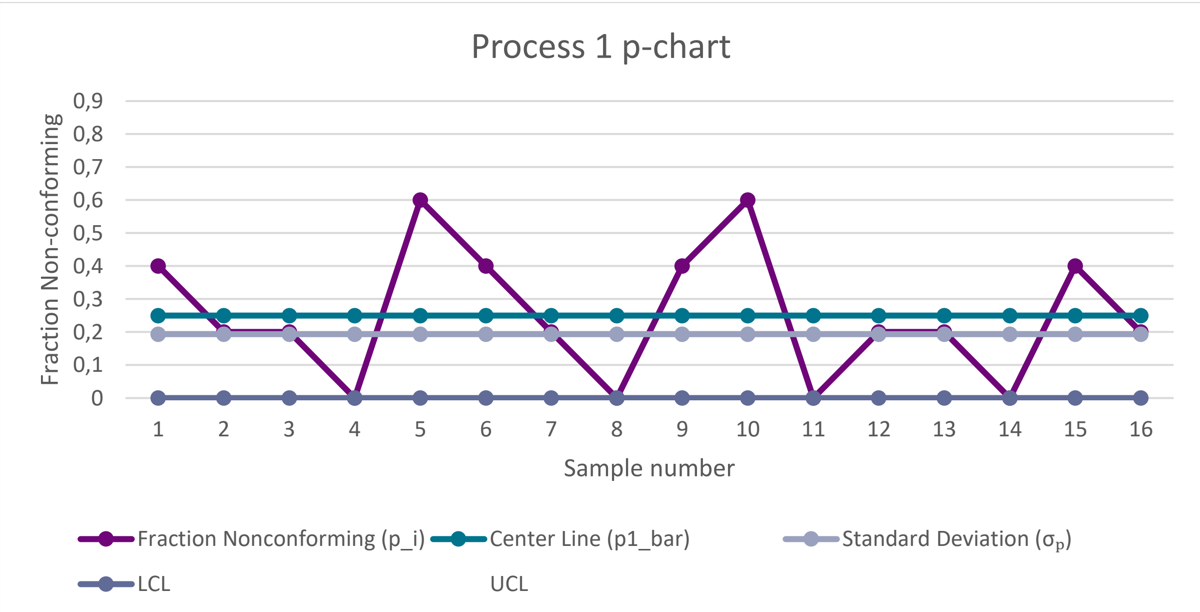
- Pareto Analysis: Helps prioritise issues by identifying the most significant sources of defects (Montgomery, 2020:49).
- Cause-and-Effect Diagram (Ishikawa): Aids in root cause analysis by mapping factors contributing to defects (Sadikin, 2023:81).
- Control Charts: Monitor process stability by distinguishing between common-cause and special-cause variation (Montgomery, 2020:44).
- Histograms: Provide a visual representation of defect distribution, supporting SPC analysis (Godina, Pimentel and Matias, 2018:7).

3. Data analysis

For process 1 :

The statistical inferences that were made showed that the data should not be assumed as a normal distribution.

he calculated LCL of -0.331 is mathematically negative due to the combination of high defect rate (25%) and small sample size (n=5). Since proportion defective cannot be negative, the LCL is set to 0.000 following standard statistical practice (Montgomery, 2019). This does not invalidate the control chart but indicates the process has only an upper control limit for practical purposes.



The capability analysis of Process 1 reveals a process mean (μ) of 0.25 and a standard deviation (σ) of approximately 0.1936. Given specification limits set between 0.05 and 0.30, the calculated Cp value of 0.215 indicates that the process exhibits substantial variability relative to the allowable range. More critically, the Cpk value of 0.086 confirms that the process is poorly centered and not capable of consistently operating within specification limits. This result, supported by a high defect rate and broad spread, implies a high likelihood of nonconformance. The process in its current state lacks the stability and precision required for a capable production system and requires both centering and variance reduction interventions.

Total Number of Non-conforming (Di_1)	20
Center Line ($\bar{p}$)	0,25
Standard Deviation (σ_p)	0,193649167
Lower Control Limit (LCL)	0
Upper Control Limit (UCL)	0,830947502
m	16
n	5
Process 1 dataset total	80

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